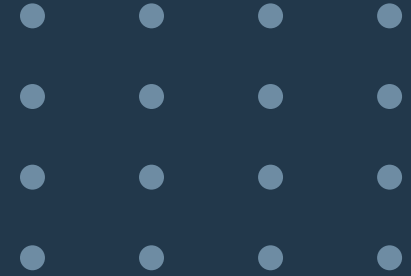


MODEL CONTEXT PROTOCOL

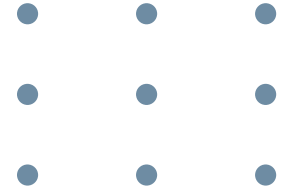


Introduction to MCP • Tracey Kroll

Tracey Kroll

Giving AI Access
to Your Data

A LITTLE ABOUT ME



ptc[®]

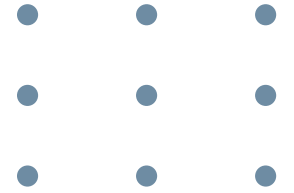


Introduction to MCP · Tracey Kroll



AGENDA

What we'll cover

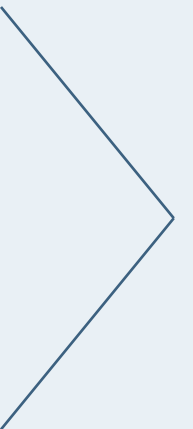
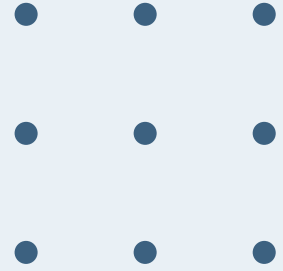


- 01 Foundations:** What MCP is, and what it is not
- 02 Architecture:** Host, client, server, and the model that decides
- 03 Consuming MCP:** Two connections, one question, no code
- 04 Authoring MCP:** Turning a stored procedure into a tool
- 05 Security:** The part the protocol does not do for you
- 06 Wrapping Up:** Takeaways, next steps, and Q&A

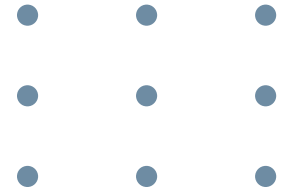
01

SECTION ONE

Foundations



THE CONTEXT GAP



Why MCP matters

My AI assistant can't see my data.

I spend half my time manually copying context to give my LLM.

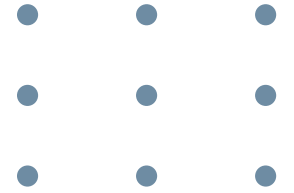
My data lives in too many systems.

It can tell me what to do, but it can't do it for me.

Every tool connects to a different set of systems.

The model isn't short on intelligence. It's short on *your* data.

ONE STANDARD, MANY SYSTEMS



MCP is...

An open standard for connecting AI applications to tools and data.

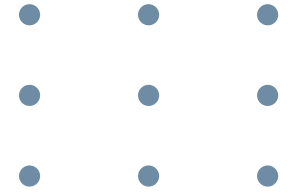
Your data stays where it is. The model reads it live.

It's bidirectional: it reads context and it **takes action**.

Any client, any server. Build it once.

The server describes itself. The model works out the rest.

NOT IN ITS JOB DESCRIPTION



MCP isn't...

Not an LLM. It doesn't reason.

Not an agent framework. Agents sit on top of it.

Not a replacement for APIs. It wraps them.

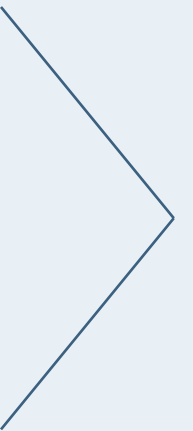
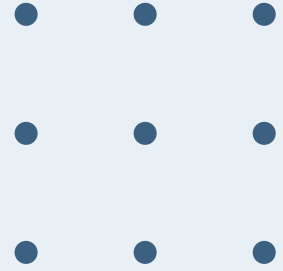
Not RAG. Nothing is copied or embedded.

Not a security model. That part is yours.

02

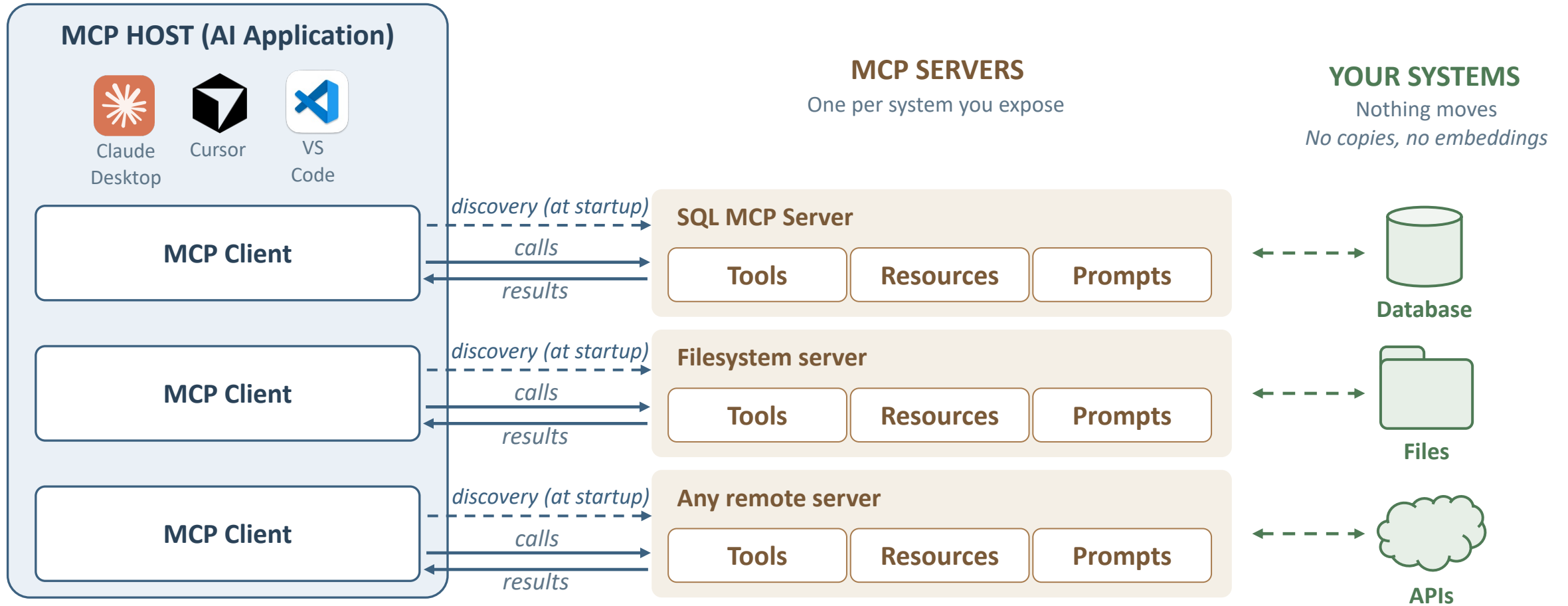
SECTION TWO

Architecture

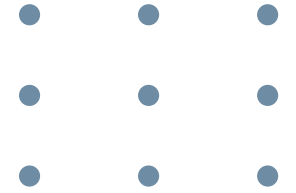


ARCHITECTURE

How the pieces fit together



THE BUILDING BLOCKS



The three MCP primitives

Everything an MCP server exposes is one of these three.

1

Tools

Executable actions the model can call — query, calculate, write.

2

Resources

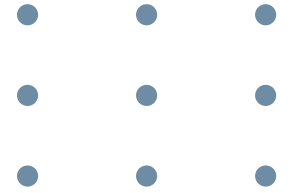
Read-only data the server hands over as context.

3

Prompts

Reusable, parameterized templates — pick one and fill in the blanks.

DISCOVERY

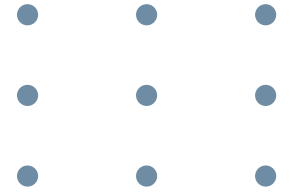


Why this isn't just an API

A server describes itself. The model reads those descriptions and decides what to do with them.

- 1 The server publishes what it can do**
A client asks `tools/list` and gets back names, typed parameters and descriptions. Nothing is compiled in.
- 2 The model chooses**
It reads those descriptions and decides which tools to call, in what order, and when it has enough to answer.
- 3 Composition happens at runtime**
Connect a second server and the model can combine both — without anyone writing integration code.
- 4 Your descriptions are the interface**
The model reasons over the names and descriptions you write. That's application design, not documentation.

RETRIEVAL



Why this isn't RAG

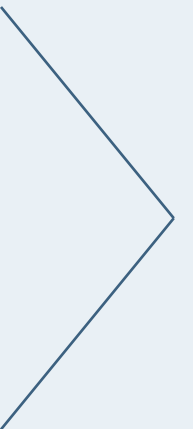
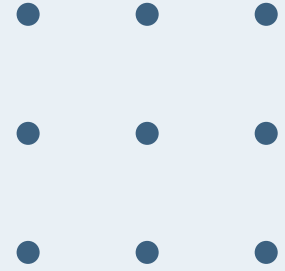
RAG copies your data into an index. MCP leaves it where it is.

- 1 Nothing is copied or embedded**
No chunking, no vector store, no index to rebuild. The data never leaves the database.
- 2 Nothing goes stale**
An index is a snapshot of the moment it was built. A tool call reads the row as it is right now.
- 3 It can act, not just retrieve**
Retrieval finds text to inform an answer. A tool issues the credit and writes the audit row.
- 4 Exact, not approximate**
Vector search returns what is similar. A query returns what matches — the same rows every time.

03

SECTION THREE

Consuming MCP: MCP as a Client



MONDAY MORNING

Meet Bay State Supply



Industrial distributor. Sam in accounts payable checks vendor invoices every Monday morning — two monitors, a highlighter, and hundreds of line items.

From the vendors

Four invoices from four vendors, no two the same.
Plus the supply contract that management signed and nobody reads.

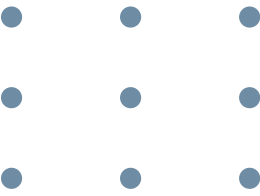


In the system

What we ordered, what arrived, what we agreed to pay, and what we have already paid.

Did these vendors bill us correctly?

WHAT THE VENDORS SENT



HOLLIS FASTENERS



HOLLIS
FASTENERS

412 Old Mill Road
Hollis, NH 03049
(603) 555-0148
ar@hollisfasteners.example.com

INVOICE

Invoice No.
Invoice Date
Your PO
Terms
Due Date

HF-2026-1184
10-Jul-2026
PO-4471
Net 30
09-Aug-2026

BILL TO

SHIP TO

Bay State Supply
Accounts Payable
1150 Providence Highway
Norwood, MA 02062

Bay State Supply
Receiving Dock B
1150 Providence Highway
Norwood, MA 02062

LINE	DESCRIPTION	QUANTITY	UOM	UNIT PRICE	AMOUNT
1	Hex Bolt 1/2-13 x 2in, Grade 5, Zinc	2,000	EA	4.1000	8,200.00
2	Flat Washer 1/2in, Grade 8, Zinc	3,000	EA	0.9600	2,880.00
Subtotal					11,080.00
Freight					0.00
Total Due (USD)					\$11,080.00

Remit to: Hollis Fasteners, PO Box 2214, Nashua, NH 03061

BRADDOCK COMPONENTS

From: Sean Whitaker <invoices@braddock.example.com>
To: Accounts Payable <ap@baystatesupply.example.com>
Date: Mon, 3 Aug 2026 09:14:22 -0400
Subject: SECOND NOTICE – Braddock Components invoice BC-77341

Hi,

Following up on the below – our records still show this as outstanding and it is now well past terms. Could someone take a look and let me know where it stands? Happy to resend backup if that helps.

Invoice number BC-77341
Invoice date August 3, 2026
Your purchase order .. P0-4462
Terms Net 30

60 ea Ball Bearing 6205-2RS @ 84.50 ea 5,070.00

AMOUNT DUE (USD) 5,070.00

Remit to:
Braddock Components
88 Furnace Street
Braddock PA 15104

Thanks,
Sean Whitaker
Accounts Receivable, Braddock Components
(412) 555-0193

SUDBURY PACKAGING

Sudbury Packaging Co.
221 Boston Post Road
Sudbury MA 01776
billing@sudburypack.example.com

INVOICE

Doc Type
Invoice

Doc No
SP-88214

Doc Date
7/2/26

Cust PO
PO-4468

Terms
Net 30

Due Date
8/1/26

Sold To
Bay State Supply
1150 Providence Highway
Norwood MA 02062

Ln	Part No	Description	Qty Shipped	UOM	Unit Cost	Ext Amount
1	SP-GLV-NBL	Nitrile Glove Blue Large	4000	EA	0.55	2200
2	SP-BOX-1212	Corrugated Carton 12x12x6 Kraft	750	EA	1.25	937.5
Subtotal						3137.5
Freight						0
Total Due USD						3137.5

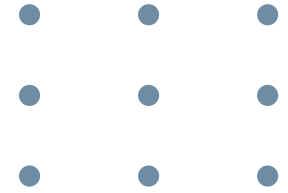
Remit to Sudbury Packaging Co PO Box 771 Sudbury MA 01776
Reference invoice number on remittance

TILTON METALS

INV_NO	INV_DT	CUST	CUST_PO_REF	TERMS	LN	PART	DESCR	QTY	UOM	PRICE	AMT
TM-5590	17/07/2026	BAYSTATE	PO-4475	N45	001	TM-ANG-2020-25	ANGLE STL 2X2X1/4 20FT	150	EA	46.2000	6930.00
TM-5590	17/07/2026	BAYSTATE	PO-4475	N45	002	TM-TUB-1000-16	TUBE STL SQ 1IN 16GA 24FT	60	EA	58.4000	3504.00
TM-5590	17/07/2026	BAYSTATE	PO-4475	N45	003	MISC-FSC	FUEL SURCHARGE 3.0 PCT	1	LOT	312.0000	312.00

USING AN MCP SERVER

Let's ask it



All four invoices have discrepancies: \$6,162.00 is billed above what we owe.

Invoice	Discrepancy	Billed	Should be	Over
BC-77341 Braddock	Duplicate of BC-77120 (same PO-4462, \$5,070), already paid 6/30/26, check 104882; PO closed (terms §7.3)	5,070.00	0.00	5,070.00
HF-2026- 1184 Hollis	Hex bolt at \$4.10 vs PO/contract \$3.85 in force on order date 6/18 (§3.2)	11,080.00	10,580.00	500.00
SP-88214 Sudbury	Gloves at \$0.55/EA vs PO 40 CS × \$48 (\$0.48/EA) (§4.3); cartons match but neither line states the CS→EA conversion (§4.2)	3,137.50	2,857.50	280.00
TM-5590 Tilton	\$312 fuel surcharge; prohibited outright (§5.2), deduct from payment (§5.4)	10,746.00	10,434.00	312.00

All quantities match the receipts, and the terms match the POs.

Create a spreadsheet



Reply



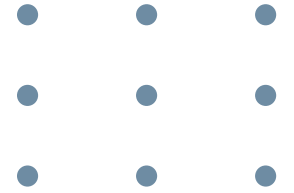
Claude is AI and can make mistakes.

Opus 5.5 High Manual

The model read four invoices in four formats, checked them against the purchase orders and receipts, and found four discrepancies. None of that data was in the prompt.

<https://youtu.be/3RmgWY6DRUc>

THE CLIENT CONFIG



Your entire job

Nothing was built here. This file just says which servers to connect to.

```
{
  "mcpServers": {
    "baystate-sql": {
      "command": "npx",
      "args": ["-y", "mcp-remote",
        "http://localhost:5000/mcp"]
    },
    "baystate-files": {
      "command": "npx",
      "args": ["-y",
        "@modelcontextprotocol/server-filesystem",
        "/path/to/invoices"]
    }
  }
}
```

1

What to call it

The name you'll see in the client.

2

How to start it

The client launches the server itself.

3

Second server, same shape

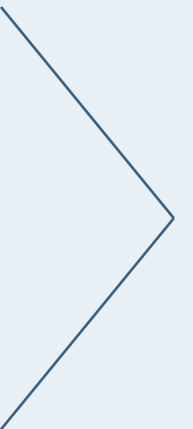
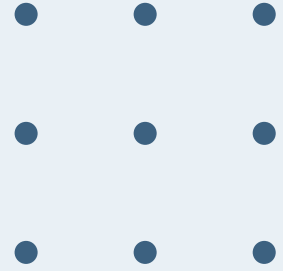
What to call it, how to start it.

Two entries and a restart. That is the whole consumer story.

04

SECTION FOUR

Authoring MCP: MCP as a Server



MEANWHILE, IN CUSTOMER SERVICE

Back at Bay State Supply

Two customers have written in with a complaint. Dana in customer service can help them, but something has to record the credit.



What Dana has to work out

The order history says one thing, the emails say another. She has to reconcile them, then apply the policy: does this qualify, and for how much?

What has to happen next

A credit gets applied according to the policy. An audit row records the credit, who issued it, and under what rule.

Reading the DB was the easy part. This time we're writing.

FROM THE DESK OF ALICIA MORROW...

From: Alicia Moreau <amoreau@graniteridgemech.example.com>
To: Customer Service <service@baystatesupply.example.com>
Date: Mon, 3 Aug 2026 08:41:17 -0400
Subject: Account C-10041 – two orders in a row

Hi,

I don't like making this kind of call, so I'll put it in writing instead.

We've had two bad orders back to back now. The June order came in with two of the six enclosures crushed – your team sorted that out quickly and I have no complaint about how it was handled. Then the July order showed up more than a week after the date we were given, and nobody called to tell us it was going to be late. We had a crew scheduled around that delivery and had to send them home.

We've bought from Bay State since 2019 and this isn't typical, which is why I'm raising it rather than just moving on.

I'd like a credit on the account, particularly for the July order and the day we lost on it. Can someone come back to me this week?

Alicia Moreau
Purchasing, Granite Ridge Mechanical
(978) 555-0142

AND ALSO, DOUG FARROW

From: Doug Farrow <dfarrow@pembertonfacilities.example.com>
To: Customer Service <service@baystatesupply.example.com>
Date: Wed, 5 Aug 2026 16:52:04 -0400
Subject: RE: RE: Account C-10077 – THIRD problem this year

This is now the third order in a row that has gone wrong and I am done being patient about it.

March – the shipment arrived with a damaged carton. I raised it and was told it was too late to do anything. Too late? It's your product and it arrived damaged.

May – an entire pallet came in water damaged. You did credit that one, I'll give you that, but it should never have happened in the first place.

July – the order was five days late. Five days. I had contractors on site waiting on that delivery and I had to pay them to stand around. I don't particularly care whose fault the address was, the fact is the material wasn't here when your paperwork said it would be.

Three orders, three problems. At some point this stops being bad luck and starts being a pattern. I've been buying from you people since 2021 and I am now seriously looking at what else is out there.

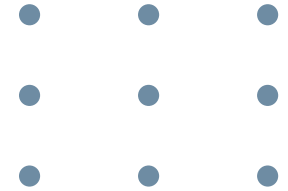
So here is what I want. Put a credit on this account and make it a real one. Three failed deliveries and a crew standing around on site is not a fifty dollar gesture and I won't accept being fobbed off with one. Make this right or tell me plainly that you won't, and I'll take it from there.

I'd like an answer this week.

Doug Farrow
Facilities Director, Pemberton Facilities Group
(508) 555-0177

WRITING THROUGH AN MCP SERVER

Two customers, one policy



Now handle the request from Doug Farrow.

Baystate service: Account standing >

Doug Farrow (Pemberton Facilities Group, C-10077) does **not** qualify for a credit, so I haven't issued one.

- **Why:** policy needs 2+ qualifying issues in the last 3 completed orders; his account shows only 1 (affected value \$840.00).
- **Likely reasons, unconfirmed:** he says the March damage was reported too late, and the July delay involved an address issue that may not count against Bay State. May was already credited.
- **Override:** only a supervisor can override eligibility, and it requires a written reason. This login is a support rep.

How do you want to handle it? I can draft a reply that declines plainly and explains why, or route it to a supervisor for an override review.

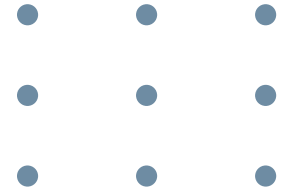


The model read Doug's complaint, checked his order history, and found he doesn't meet the policy. No credit was issued.

Support rep role:
<https://youtu.be/zjK1zZsGBtc>

Supervisor role:
<https://youtu.be/QOjE7OaUhGc>

AUTHORING AN MCP SERVER



Three steps, no new logic

Nothing here decides anything. The decisions were already in the database.

WHAT YOU ALREADY HAVE



The stored procedure

The policy and the audit row already live in `usp_IssueServiceCredit`, not in a prompt.



The authorization model

The procedure reads the login. The model never gets more than the role already had.

WHAT YOU ACTUALLY DO



Expose the procedure as a tool

One `dab add` command. `--mcp.custom-tool` registers it by name.



Describe the parameters

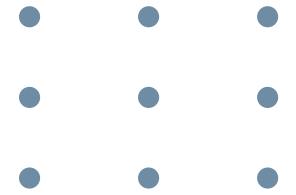
What the model reads to call it correctly. Written in the same config file.



Point the client at the server

One entry in a Claude configuration file. Restart.

EXPOSING AN MCP SERVER



Configuration, not code

The stored procedure was already there. This is what was added to make it a tool.

```
"IssueServiceCredit" : {  
  "source" : {  
    "object" : "service.usp_IssueServiceCredit",  
    "type" : "stored-procedure"  
  },  
  "permissions" : [  
    { "role" : "bss_support_rep", "actions": [ "execute" ] },  
  ],  
  "mcp" : { "custom-tool": true }  
}
```

1

Which procedure

The one already written and tested.

2

Who may run it

The roles you already manage.

3

Register as a tool

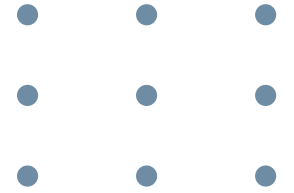
One line. Now an agent can find it.

No pipeline. No ETL. No project plan.

THE ECOSYSTEM

Or don't build one at all

Hundreds already exist.



Filesystem

GitHub

Email

Fetch

SQL Server

Memory

Slack

Brave Search

Notion

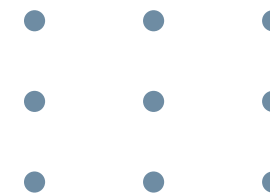
Stripe

Playwright

+ hundreds more

Prefer not to write one at all? n8n, Zapier and Make will expose a workflow as a server.

WHAT MAKES A GOOD MCP SERVER?



Clear beats clever

How the model picks it

Single-responsibility tools

One clear outcome. Solve a specific business problem or workflow.

Standardized, clear naming

Avoid generic names. Stay consistent across the server.

Rich descriptions

Treat the description as UI for the model.

What it gives back

Predictable in and out

Well-defined parameters, consistent results.

Return the reasoning

Send back the rule applied and the limit, so the caller can explain itself.

Helpful errors

Enough detail for the model to understand and recover.

When it retries

Safe to call twice

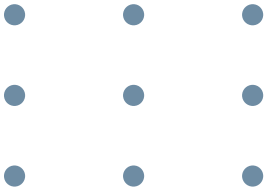
A model will retry — after a timeout, or just to be sure.

Design for it: the same call twice should leave the database in the same state.

Two calls must not mean two credits.

If you have written a good stored procedure, you've already done the work.

MCP VS LLM GENERATED SQL



Better answers, less work

Both can return a correct number. Only one of them arrives at it the same way

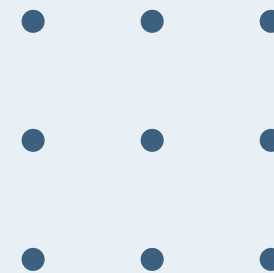
	LET THE MODEL WRITE THE SQL	GIVE IT A TOOL YOU BUILT
How it knows your schema	You paste it into the prompt, and keep it current.	The server describes itself.
The same question, asked twice	Two different queries. Two chances to be wrong.	The same tool call. The number is computed the same way.
Who wrote the logic	The model, at runtime, from whatever it was told.	Your team, in a proc, reviewed and tested.
What it can reach	Whatever the login can reach.	Only the entities you configured.
When the rule changes	Update the prompt and hope.	Change it in one place. Everything follows.
When it is wrong	A confident answer nobody checks.	A guardrail that raises an error.
Who is accountable	Nobody can say what ran.	An audit row with who, what and when.

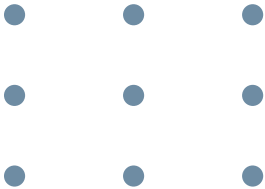
Microsoft leaves natural-language-to-SQL out deliberately: the queries you'd most want generated need the most scrutiny.

05

SECTION FIVE

Security





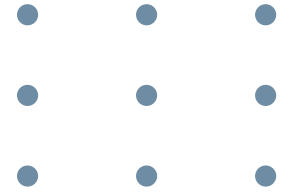
Where the data goes

The honest answer to the question everyone asks after a demo like this.

Your database	Stays exactly where it is. Nothing is copied, exported or synced.
What does leave	Whatever the model reads: query results, file contents, the text of a customer email. Same as pasting it into a chat window.
Personal ChatGPT or Claude	Conversations train the model unless the setting is turned off. At time of publication Anthropic retains for five years if it is on. Check your AI provider’s privacy settings
Team or Enterprise	Usually covered by Commercial Terms. Not used for training. This is the account type to use for work.
API	Also USUALLY excluded from training. Billed per token rather than per seat.
If you are regulated	Ask about Zero Data Retention and a BAA. Both are Enterprise features.

A personal subscription used for company work is the thing to fix first.

SECURITY



You still have to design a safe server

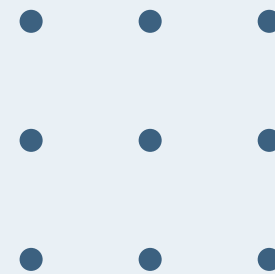
Design the server as though the model is an untrusted client — because it is.

- 1 Give the MCP server a dedicated login whose role only grants what it needs.
- 2 Expose only the tables, views, procedures and operations it needs.
- 3 Begin with read-only capabilities.
- 4 Require approval for consequential operations.
- 5 Log the tool name, arguments, user identity, duration and outcome.

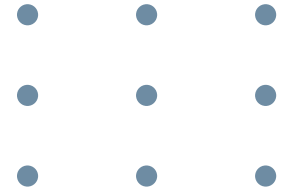
06

SECTION SIX

Wrapping Up



TAKEAWAYS



The short version

MCP standardizes how an AI application connects to tools and data.

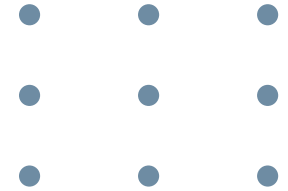
A server publishes what it can do. The model decides what to call.

It works with what you already run, and the skills you already have. No new language.

Access should be intentional, limited and secure. That part is yours, not the protocol's.

Start small. One useful connection unlocks a lot.

AGENTS



This is what agents run on

An agent is what you saw today, with something other than a person starting it.

What you saw today

A person asked. The model worked out which tools it needed, read what it had to read, and stopped where the database told it to stop.



What changes

A schedule, an inbox, a queue. Nobody types anything, and the tools do not change at all.

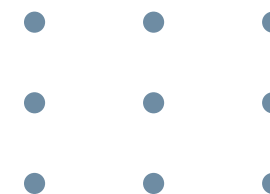


What that needs

Rules nobody can talk their way past, and a record of what ran. You have already seen both.

MCP is not an agent framework. It is what an agent acts through.

WHERE TO START MONDAY



Your first MCP server

Start read-only, start small, and put it in front of one team before you scale it.

1

Pick a database and a dedicated identity

Choose one non-production database and create a read-only service account just for the MCP server.

2

Define three to five tools on paper first

Name each tool, write its description, and list the parameters and return shape before writing any SQL.

3

Expose them with configuration, not code

Install Data API Builder — one command. Then point it at your existing tables, views and procedures.

4

Test the contract in MCP Inspector

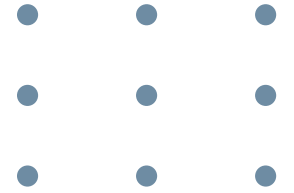
Verify the tool list and invoke each tool manually, before any AI assistant is connected to it.

5

Connect one host and pilot with one team

Wire it into VS Code or Claude Desktop, enable logging, and widen access only once you trust it.

IF YOU ARE FOLLOWING ALONG



Things that might stump you

Every one of these cost me an afternoon. All five are documented in the repository.

dab: command not found

The tool installs to ~/.dotnet/tools, which is not on the PATH by default. Add it to ~/.zprofile.

You must install or update .NET

Data API builder 2.0 targets .NET 8. A newer SDK does not satisfy it. Install the .NET 8 runtime alongside whatever you have.

No stored procedure definition found

The login needs VIEW DEFINITION. EXECUTE lets it run the procedure but not read the signature, which is what DAB needs.

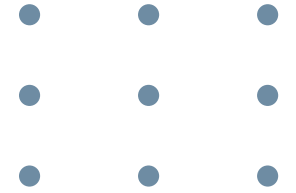
Not valid MCP server configurations

Claude Desktop only accepts stdio servers. Wrap an HTTP endpoint with mcp-remote rather than giving it a url.

Parameter descriptions come out shifted

--parameters.description splits on commas with no escaping. Edit dab-config.json by hand.

EVERYTHING IS IN THE REPO



Your turn



- Both demos — schema, data, configs
- Step-by-step READMEs, including what broke
- These slides

`github.com/traceykroll/
IntroductionToMCP`

My code, open sourced under MIT. Slides may be shared with attribution.

THANK YOU!

Questions?

Tracey Kroll

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Slides, demos and walkthroughs

[github.com/traceykroll/
IntroductionToMCP](https://github.com/traceykroll/IntroductionToMCP)

